

01 / BUSINESS IDEA



What is Ghost Layer?

Imagine a Formula 1 race car. The car is super fast, but if pit stops take too long or the driver steps on the wrong pedal, it loses the race!

💡 **The ELI5 Concept:** Ghost Layer is an **Invisible Smart Mechanic** for computer supercomputers (GPUs). It jumps inside, clears traffic jams in the computer brain, and makes AI run **up to 2x faster!**

+65.6%

LLM Speedup

10,000+

Auto-Scaled GPUs

\$0 Risk

Instant Rollback

02 / HOW IT WORKS

What Does Ghost Layer Actually Do?

Think of training an AI like baking 1,000,000 cookies. If you use a tiny spoon and check the recipe every 5 seconds, it takes 100 years!

4 Simple Steps:

1. **Observe:** Sees where the computer is waiting.
2. **Diagnose:** Finds the traffic jam.
3. **Safe-Apply:** Makes the oven twice as fast.
4. **Verify:** Tastes the cookie to make sure it's not burned!

03 / SCALABILITY



How Do We Scale It?

Managing 10,000 computers normally takes hundreds of engineers.
Ghost Layer scales automatically!

🔧 **Plug & Play:** Engineers add **one single line of code**. Ghost Layer automatically clones itself across 1 or 10,000 GPUs instantly without human intervention!

1 Line

Integration Hook

0 Sec

Manual Setup

100%

Autonomous Fleet

04 / THE PROBLEM



Why Companies Waste Millions

Big AI companies rent supercomputers for **\$30 to \$100 every single hour** per chip!



The Hidden Problem: Supercomputers spend 40% of their time sitting idle waiting for memory to arrive down a slow wire. Ghost Layer clears the highway so GPUs never sit idle!

05 / SECRET SAUCE



The Ghost Layer Secret Sauce

Why is it called "Ghost"? Because it works **invisibly in the background** without restarting training!

🧩 3 Secret Ingredients:

- **Closed-Loop Tuning:** Tries speed boosts live.
- **Mathematical Guardrails:** Guarantees the AI doesn't get dumber.
- **1-ms Rollback:** Rewinds time if anything goes wrong.

06 / SAFETY

Safety First: Keeping the AI Smart

Tuning GPUs without safety is like driving with broken brakes. Pushing too hard can ruin an AI model!

 **Loss Trajectory Guardrails:** Ghost Layer tracks the exact learning math. If loss diverges, it instantly revokes auto-apply rights and rolls back in 1 millisecond. Zero ruined training runs!

07 / HARDWARE BASICS



How GPUs Think (Simplified)

Understanding CPUs vs. GPUs using Minions!



CPU vs. GPU:

- **CPU:** 1 super genius professor. Solves hard math one by one.
- **GPU:** 10,000 Minions painting together! Ghost Layer makes sure all 10,000 Minions keep painting without stopping.

08 / VRAM MAGIC



VRAM Memory Magic Tricks

VRAM is the GPU's backpack. Stuffing too many toys inside breaks the zipper (Out Of Memory)!

🔮 Memory Magic:

- **Precision Tuning:** Uses 16-bit numbers (takes half the space).
- **Memory Layout:** Eliminates empty gaps.

Result: Train giant AI models on cheaper GPUs!

09 / EXECUTION SPEED



Training AI in Record Time

Finishing 4 hours of homework in 2 hours means more time to play outside!

⚡ Empirical Speedups:

- **Native LLM:** Jumped from 65.4k to 190.6k tok/s (+65.6%).
- **ResNet-50 Vision:** +19.8% speedup.

Finishing training days earlier saves millions in rental costs!

10 / CFO RECEIPTS

Saving Real Money & Printing Receipts

CFOs don't want promises—they want verifiable financial receipts!

 **Audit Receipt:** Ghost Layer computes exact GPU-hours saved, baseline vs. optimized costs, and prints verified dollar receipts for the finance team.

11 / SECURITY



Privacy & Security: Local-First

Companies never want their secret AI recipes leaving their building.



Local-First Security: Ghost Layer runs entirely inside customer hardware. Zero data exfiltration. It reads hardware telemetry, NEVER private AI text or customer data!

12 / BUSINESS MODEL



How We Make Money

We use a zero-risk **Performance Share** pricing model!



Value Split: We take a 25% slice of the actual money we save the customer. Customer keeps 75%. If we save \$0, they pay \$0!

13 / COMPETITIVE MOAT

Why Big Chip Makers Can't Stop Us

Why can't NVIDIA or PyTorch just replace Ghost Layer?

 **Knobs vs. Drivers:** NVIDIA builds the engine and 500 dashboard knobs. Ghost Layer is the **Autonomous Race Car Driver** making live decisions inside the race loop!

14 / BENCHMARK PROOF

Real-World Exceptional Results

Empirical evidence measured directly on physical GPUs:

190.6k

LLM tok/s (vs 65.4k
baseline)

+22.99%

250-Cluster Fleet
Projection

100%

Loss Divergence Safe

15 / FUTURE VISION

The Future Vision

A world where every GPU on Earth runs at 100% peak efficiency!

 **Impact:** Greener AI compute, half-price training costs for startups & researchers, and the universal autonomous compute layer across all clouds!  